Photonics Applications

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Photonics Applications (A11386)

Short Title: Photonics Applications
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author JHOULIHAN
Credits 5 **Level:** Advanced

Description of Module / Aims

This module introduces a variety of applications in photonics.

Programmes

PHYI-0033	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	4 / 8 / M

Indicative Content

- Einstein relations, optical feedback and threshold conditions, resonator modes
- Laser types: gas, solid-state, semiconductor, quantum well, vertical cavity, and parametric lasers
- Frequency stabilisation, distributed feedback structures
- Mode locking, Q switching
- Optical waveguiding, propagation in optical fibres
- Dielectric waveguide, optical fibre waveguide, waveguide modes
- Fiber-optic communications and technology
- Optical fibre fabrication
- Attenuation and scattering mechanisms
- Dispersion, pulse spreading and bandwidth effects; dispersion compensation
- Optical sources and detection
- Light emitting diodes and laser diodes: structures and characteristics; modulation schemes
- Photomultipliers, p-n, pin and avalanche photodiodes, quantum efficiency, response speed, bandwidth; noise issues

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Interpret the theory underpinning advanced photonic devices.
2. Design and evaluate a range of photonic systems used in communications and metrology.
3. Design and test optical communications networks.
4. Design and manage various photonic systems.

Learning and Teaching Methods

- Lectures.
- Practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2
Continuous Assessment	30%	
Practical	30%	3,4

Assessment Criteria

<40%: Inability to understand the basic concepts encountered in this module.
40%-49%: Ability to demonstrate a basic understanding of advanced concepts in physical optics and photonic devices.
50%-59%: Ability to understand and explain advanced concepts in physical optics and photonic devices and describe some applications of photonic systems.
60%-69%: All the above, and in addition, apply problem-solving techniques to problems in photonics. Describe some key optical applications of photonic systems used in communications and metrology.
70%-100%: All previous to an excellent level. Demonstrates an ability to comprehensively discuss the course material and analyse and design a range of photonic systems.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	18	
Independent Learning	93	

Essential Material

- Ghatak, A. and K. Thyagarajan. _Introduction to Fibre Optics_. Cambridge: Cambridge University Press, 1998.
- Hecht, E. _Optics_. New York: Addison Wesley Longman, 2000.
- Rosencher, E. and B. Vinter. _Optoelectronics_. Cambridge: Cambridge University Press, 2002.
- Wilson, J. and J. Hawkes. _Optoelectronics_. London: Prentice-Hall, 1998.

Requested Resources

- SCIENCE LAB: Physics